

MOS 6530 ROM replacement

If you are fiddling around with old 8-bit Commodore computer stuff, sooner or later you will come across a part called the 6530. It was used in the KIM-1 computer and in the dual drive IEEE disk drives produced by Commodore to name a few.

The 6530 is called a RRIOT (ROM, RAM, I/O and Timer). It contains a 1 kbyte ROM, 64 bytes of RAM, two 8-bit I/O ports (similar to a PIA or VIA) and a timer. Together with a CPU you can build a very tiny 6502 system. In my case, the problem was the ROM, as I planned on replacing the DOS V2.5 with a DOS 2.7 on an 8050 drive.

The 6530 has a sister part, the 6532. The major difference between the two is that 6532 does not have a ROM (it is called a RIOT for this reason) and twice the amount of RAM, but the timer and I/O ports are identical as far as software is concerned. By adding some logic and an EPROM it is possible to replace a 6530 with a 6532, and that was done years ago by Ruud Baltissen (see <http://www.baltissen.org/newhtm/6530repl.htm>). There is even a PCB readily available for the project on PCBway. It is of use when a 6530 has broken down.

For me this was all a bit overkill and I found the PCB to be rather large. I merely wanted a 6530, but with a different ROM contents. After obtaining the service manual for the SFD1001 it was found that Commodore had faced the same problem in the past and solved it by producing a PCB that contains a 6530 (in this case with the wrong ROM contents), one inverter and an EPROM socket. This project is effectively a recreation of that board, although I ended up with a smaller footprint.

There is one proviso. The 6530 has some mask programmable options, like the function of two I/O pins (I/O pin or CS pin; in case of the latter the polarity is also a mask option) and the internal address decoders for the RAM and I/O ports/timer. This means that, apart from the ROM contents, in practice not all 6530's are created equal. This project will only work if the function of the I/O pins PB5 and PB6, and their polarity in case they are (a) chip select pin(s) is the same, and the internal address decoders are the same. This will be true in cases where only a different ROM contents is required using a working 6530 (as is the scope of this project).

How the circuit works

The ROM on the 6530 is addressed using the 10 address lines A0 through A9 and asserting the active level for ROM access on the pin RS0 (pin 4). Its polarity is possibly a mask option, but in CBM IEEE drives it is active low. The MOS 6530 datasheet used by the author is rather vague on this point but hints at it being active low also (note: the pin name is always given without a preceding slash or stripe overhead suggesting active high).

The 6530 on the board shares 39 pins with the connector connecting it to the main board except one: pin4 (RS0). The 6530 ROM is disabled by tying pin 4 of the 6530 on the board to the +5V supply. The EPROM holding the replacement code is a 2716. It is twice the size required¹. For this reason, its address line A10 is tied to ground. The remaining address lines of the EPROM are connected to the corresponding address lines of the 6530. The same goes for the data bus. In order to guarantee read-only access, the /OE pin of the EPROM is connected to the RS0 pin of the original 6530 on the main PCB, and /CS of the EPROM receives the inverted R/W line, meaning it will only drive the data bus on read cycles when it is selected. That's all.

[Appendix A](#) contains the original Commodore schematic diagram.

¹ Single 5V byte wide EPROMs with a capacity of 1 kbyte do exist (Intel i2758 and TI TMS2508 for example) but are impossible to find nowadays.

Some small variations

The final circuit (see [Appendix B](#)) is slightly different from Commodore's circuit. The first thing noticed was that the inverted R/W line is connected to the /CS pin of the EPROM, and the RS0 line of the 6530 to the /OE pin. Although this works, this has the disadvantage that the EPROM will only enter lower power standby mode during write cycles, causing a needless waste of power. The access times of 2716 based on /CS and/or /OE are the virtually the same in practice and sufficient in this case. So the /CS and /OE pins were swapped.

Apart from this, the connection to A10 of the EPROM is tied to the supply voltage instead of ground. This is cause for the code to reside in upper half of the EPROM, which is more logical as the CPU fetches its vectors from there. For the rest, EPROM was left connected the same way. This gives the option to use a 2732(A), provided the ROM contents is programmed in the upper 1kbyte of the EPROM as well, or programmed four times across the entire address area.

The second thing that was changed is the single inverter, leaving the other five in the package unused. The inverter function can also be obtained using an NAND, NOR or XOR gate, giving the option of creating some additional logic using the leftover three gates. This may come in handy in case the polarity of the /CS pins of 6530 does not match but the address decoders (kargely) do, in order to add some simple additional address decoding is needed (for this reason, the three unused gates of it were left unconnected on purpose). The chip chosen was the 74LS00 (4 NAND gates). This was needed, as the EPROM must only be selected (/CS and /OE pin both driven low) when PHI2 is high. This requirement stems from the fact that in the floppy drive two CPU's are active, each using a half of the PHI2 clock. Not ANDing PHI2 into the /OE or /CS pin will cause a bus conflict between the two CPU's.

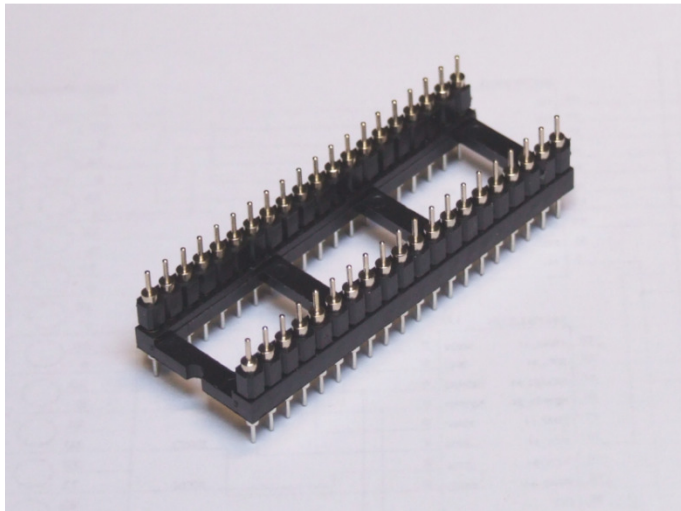
Building the circuit

Building the circuit using the PCB designed is done the easiest in the following way:

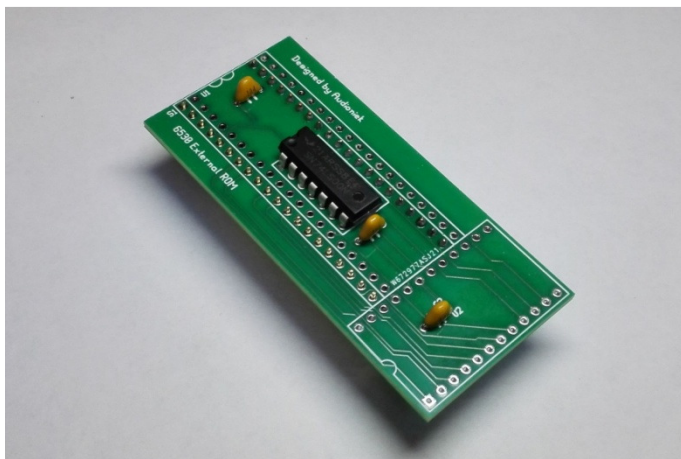
Mount the three decoupling capacitors;

Mount U3 (a 74LS00). Do not use a socket for it;

If you bought the needle pin strip as one row of 40 pins, cut it in half. Use the 40 pin DIL socket for the 6530 as a guide/template by inserting the thin needle ends of the two 20 pin needle strips into it. Work carefully as the needles are very thin and bend or break off easily:

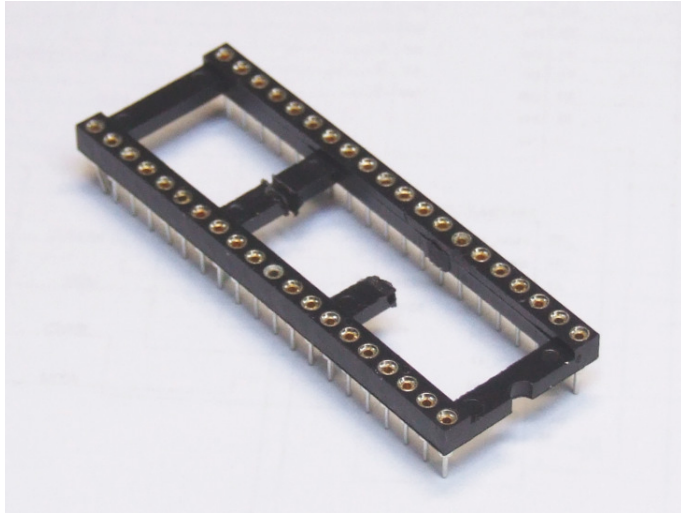


Insert the resulting package into the holes on the PCB intended for S1 on the solder side. Solder the strips on the component side of the PCB. This approach will ensure the correct spacing between the two rows of needle pins:



After soldering, carefully remove the 40 pin DIL socket from the needles.

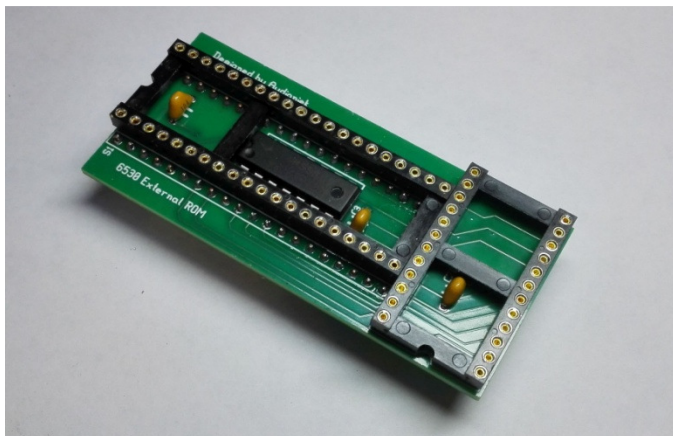
Most 40 pin DIL sockets have bridges in the middle. These unfortunately interfere with the 74LS00 sitting below the 6530, so they have to be removed. Do not just cut them away with cutting pliers as this will probably snap the socket in two. Use an old but still sharp knife and heat it using a flame or your soldering iron. Use the heated knife to cut the bridges in the socket in the middle, and then cut them out of the socket with cutting pliers:



Then solder the now bridgeless socket into the holes intended for U1 (the 6530) on the PCB. Take care the notch is in the proper place. Soldering the socket is done from the soldering side of the PCB.

Finish the build by mounting the socket for the EPROM U2.

The finished product should look like this:



Using the circuit

Putting the circuit to work is pretty straightforward. First program a 2716 or 2732(A) EPROM with the ROM content needed/desired. Remember to put the contents in the upper 1k of the EPROM, or completely fill it with 2 or 4 equal copies of the ROM content.

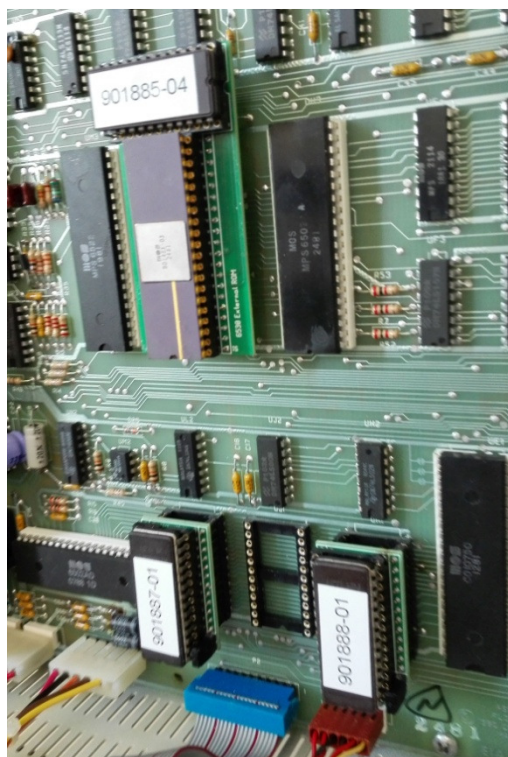
Insert the programmed EPROM into the socket intended for it.

Remove the 6530 from its original socket, and insert in the socket provided for it on the PCB.

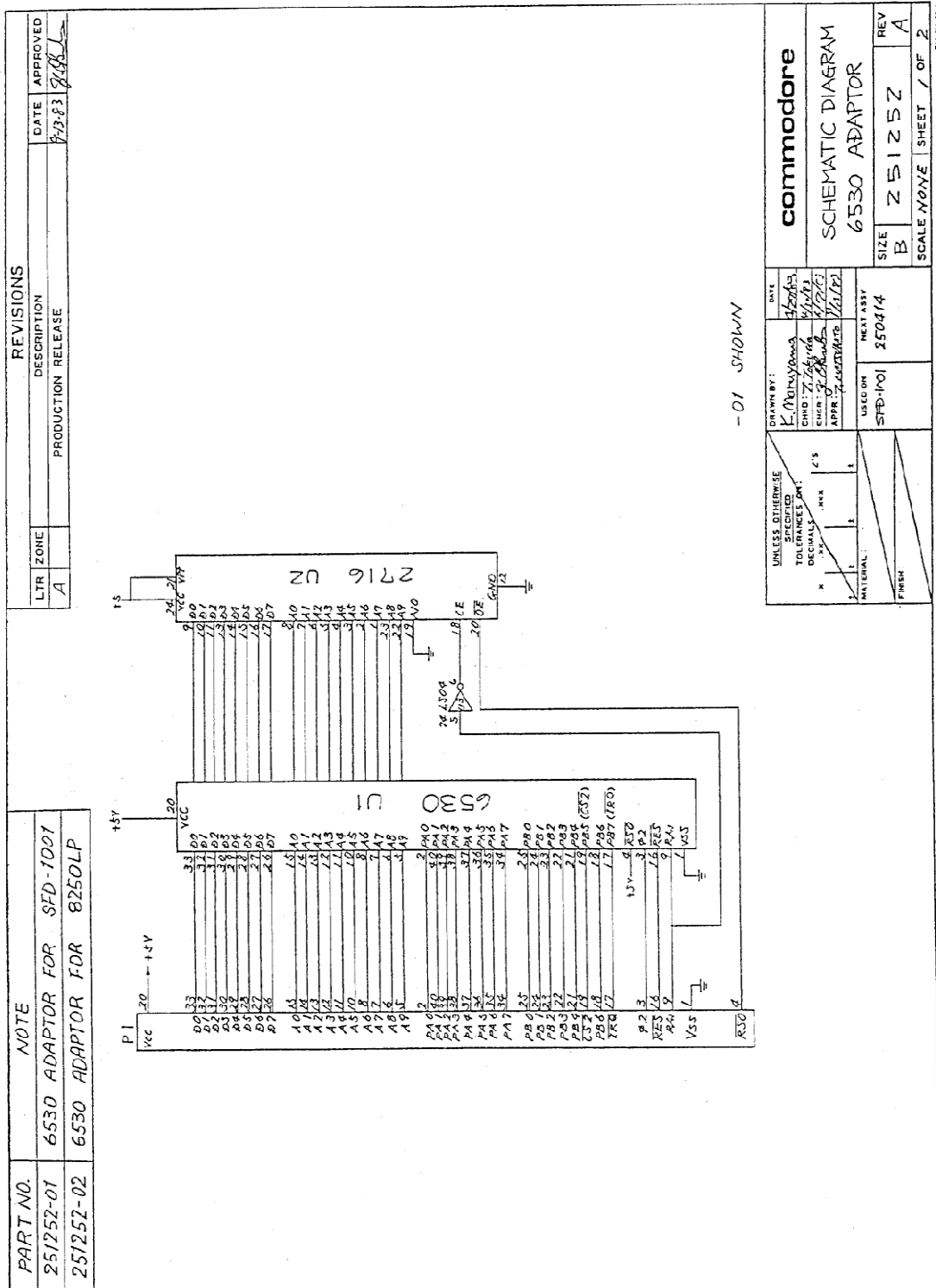
Put the PCB in the socket where the 6530 originally was. The notch indicating pin1 on the 6530 should be on the same side as it originally was.

That's it. Enjoy the result.

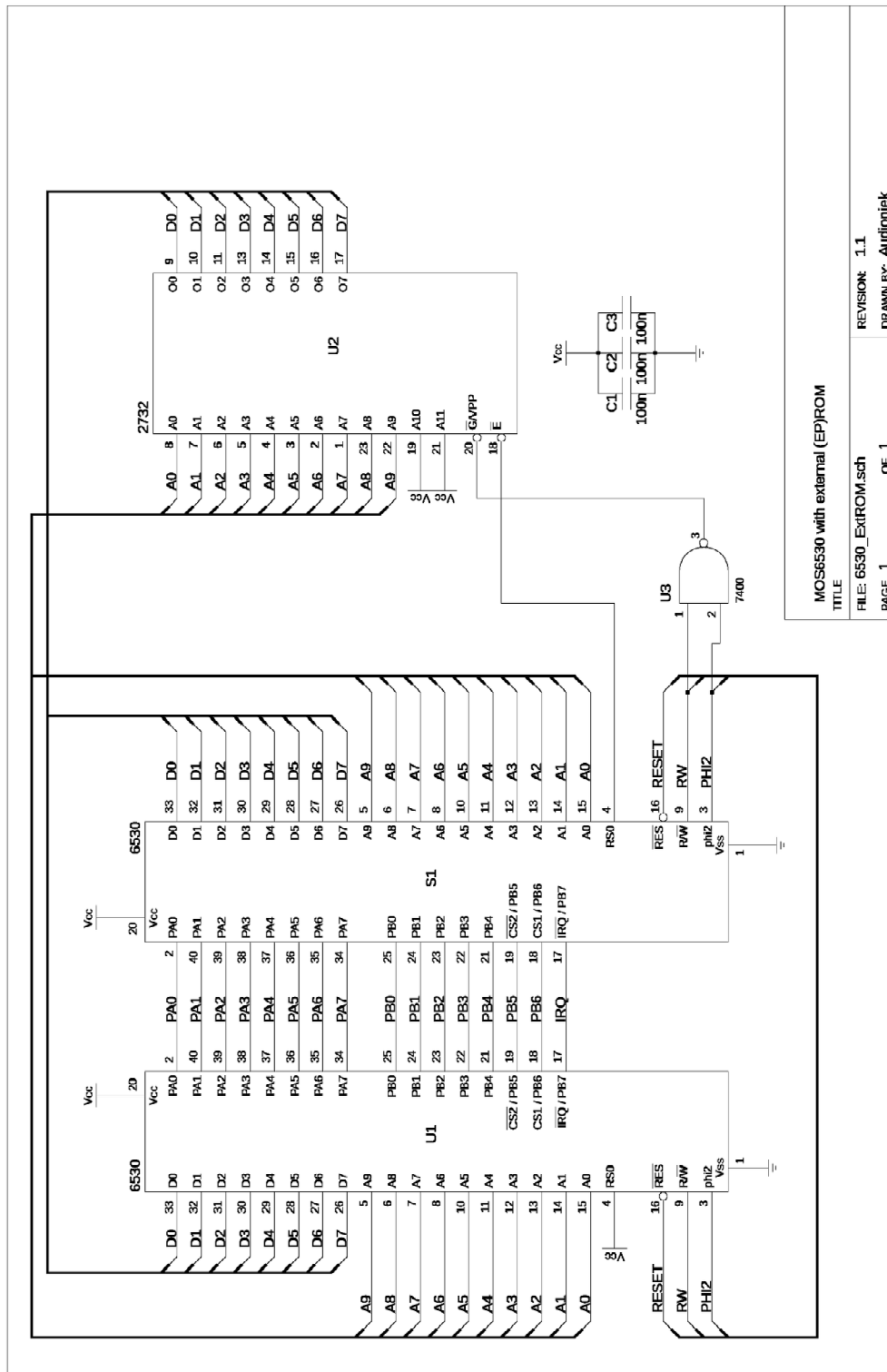
Here are two photos of the finished product in CBM 8050 drives converted to DOS V2.7:



Appendix A: the original Commodore circuit



Appendix B: schematic diagram



Appendix C: bill of materials

Reference	Description	Qty
U1	40 pin 600mil wide IC socket	1
U2	2716, 2732 or 2732A EPROM, 450ns or faster	1
U3	74LS00 (DIL package)	1
S1	Pin strips for IC socket, 2.54mm spacing	2
C1-3	100nF ceramic capacitor, 5.08 mm spacing	3
-	24 pin 600mil wide IC socket for U2	1
-	PC board	1

Appendix D: picture of PCB

